

Improving Tip-of-the-Tongue Retrieval with Transformer-Based Reranking

Phase 2: Experiment Tracking, BM25 Baseline, and Preliminary Reranker Results

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Abstract—This report presents the second phase of an ongoing investigation into transformer-based reranking for Tip-of-the-Tongue (ToT) retrieval. Building on the exploratory analysis of Phase 1, we introduce a fully reproducible two-stage retrieval pipeline: a BM25 first-stage retriever tuned via a Weights & Biases parameter sweep, followed by a cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`). All experiments are conducted on the TREC-ToT 2025/dev1 split (142 queries, same 2024 corpus). BM25 achieves $RR@10 = 0.0751$ and $nDCG@10 = 0.0812$. The cross-encoder reranker scores $RR@10 = 0.0463$ and $nDCG@10 = 0.0635$, falling below the BM25 baseline; a Wilcoxon signed-rank test confirms this difference is not statistically significant ($p = 0.1023$). A key finding is that all 142 queries have zero Jaccard keyword overlap with their top-ranked BM25 result, confirming the purely semantic nature of ToT queries and motivating stronger semantic models for future work.

Index Terms—information retrieval, tip-of-the-tongue, BM25, cross-encoder reranking, dense retrieval, bi-encoder, TREC, Weights & Biases, Wilcoxon test

I. INTRODUCTION

Phase 1 of this project established that the TREC Tip-of-the-Tongue 2024 dataset poses exceptional challenges for lexical retrieval systems: queries average 238 words, documents average over 6,700 words, and the vocabulary gap between descriptive queries and formal Wikipedia text is large. These observations motivated a two-stage architecture in which BM25 produces an initial candidate set that a semantic model then reranks.

Phase 2 realises this architecture end-to-end. An important observation from the TREC-ToT task design is that *all* queries are by construction long and narrative, and *all* have near-zero lexical overlap with relevant documents. Treating query length and keyword overlap as independent subgroup variables would therefore collapse into the same experiment. We instead define three distinct research questions that each isolate a different variable:

- **RQ1:** Does cross-encoder reranking significantly improve retrieval effectiveness over a BM25 baseline for Tip-of-the-Tongue queries (Wilcoxon signed-rank test, $\alpha = 0.05$)?

- **RQ2:** Is the reranker’s gain modulated by query length (within the ToT population), given that longer queries suffer greater truncation under the 512-token cross-encoder limit?
- **RQ3 (Phase 3):** Does replacing the BM25 first stage with dense bi-encoder retrieval improve effectiveness on a task where lexical matching is structurally insufficient?

RQ3 is scoped to Phase 3; this report establishes the BM25 and cross-encoder baselines that RQ3 will be measured against.

II. DATASET AND EVALUATION SETUP

A. Corpus and Queries

The TREC Tip-of-the-Tongue 2024 corpus contains 3,185,450 English Wikipedia documents. We use the `trec-tot/2025/dev1` query set (the same corpus), which provides 142 queries with publicly available relevance judgements (qrels). The 2024 test split’s qrels are not publicly released; the 2025/dev1 split was therefore used as the evaluation set throughout all experiments. All data is accessed via the `ir_datasets` library [9].

B. Metrics

We report three standard metrics:

- **RR@10** (Reciprocal Rank at 10, mean = MRR@10): captures whether the top-10 results contain at least one relevant document and how highly it is ranked.
- **nDCG@10:** measures the quality of the full top-10 ranking.
- **R@100:** recall of all relevant documents in the top-100, which reflects first-stage retrieval coverage.

All metrics are computed with PyTerrier’s `pt.Experiment` backed by `ir_measures` [10].

III. EXPERIMENT TRACKING WITH WEIGHTS & BIASES

All experiments are logged to the public Weights & Biases project `is584-tot-retrieval`¹. WANDB tracks hyperparameters, metrics, and run metadata, enabling full reproducibility and comparison across runs.

¹<https://wandb.ai/ilaydaergur/is584-tot-retrieval>

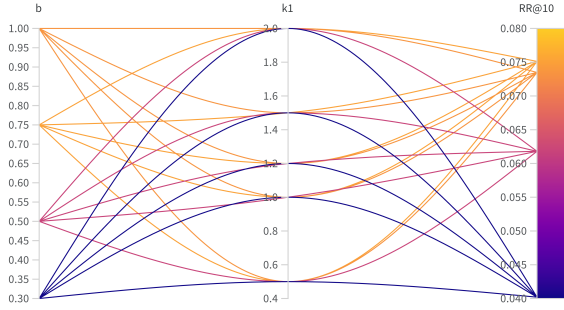


Fig. 1. WANDB parallel-coordinates plot of the BM25 grid sweep (20 runs). Each line represents one (k_1, b) configuration; colour encodes RR@10 (yellow ≈ 0.080 = best, dark blue ≈ 0.040 = worst). Runs with $k_1 \in [1.2, 1.5]$ and $b = 0.75$ consistently achieve the highest scores.

TABLE I
TOP-5 BM25 CONFIGURATIONS FROM THE WANDB SWEEP

k_1	b	RR@10	nDCG@10
1.2	0.75	0.0751	0.0812
1.0	0.75	0.0737	0.0799
1.5	0.75	0.0730	0.0805
1.2	0.50	0.0718	0.0790
2.0	0.75	0.0714	0.0798

A. BM25 Parameter Sweep

A grid sweep was run over BM25 hyperparameters: $k_1 \in \{0.5, 1.0, 1.2, 1.5, 2.0\}$ and $b \in \{0.3, 0.5, 0.75, 1.0\}$, totalling 20 runs. Figure 1 shows the parallel-coordinates visualisation from WANDB; Table I lists the top-5 configurations by RR@10. The best configuration found was $k_1 = 1.2$, $b = 0.75$, which is also the commonly recommended BM25 default and was used for all subsequent experiments.

IV. BM25 BASELINE

A. Implementation

The BM25 index was built with PyTerrier’s `IterDictIndexer` over the full 3.18M-document corpus. Two fields are indexed (text, title); a 2,000-character text snippet is stored as document metadata so the reranker can score (query, document) pairs without a second corpus pass. Retrieval returns up to 100 candidates per query.

B. Results

Table II reports aggregate metrics for the BM25 baseline. With $R@100 = 0.218$, approximately 22% of relevant documents appear in the top-100 candidates, providing the reranker’s ceiling. The RR@10 of 0.075 reflects the inherent difficulty of the task: most queries receive no relevant result in the top 10.

V. PRELIMINARY RERANKER RESULTS

A. Model and Implementation

The cross-encoder model `cross-encoder/ms-marco-MiniLM-L-6-v2` was used to score all (query, document)

TABLE II
AGGREGATE RETRIEVAL METRICS ON TREC-TOT/2025/DEV1 (N=142 QUERIES)

System	RR@10	nDCG@10	R@100
BM25 ($k_1=1.2$, $b=0.75$)	0.0751	0.0812	0.2183
CrossEncoder reranker	0.0463	0.0635	0.2183
Δ (Reranker – BM25)	−0.0288	−0.0177	0.0000

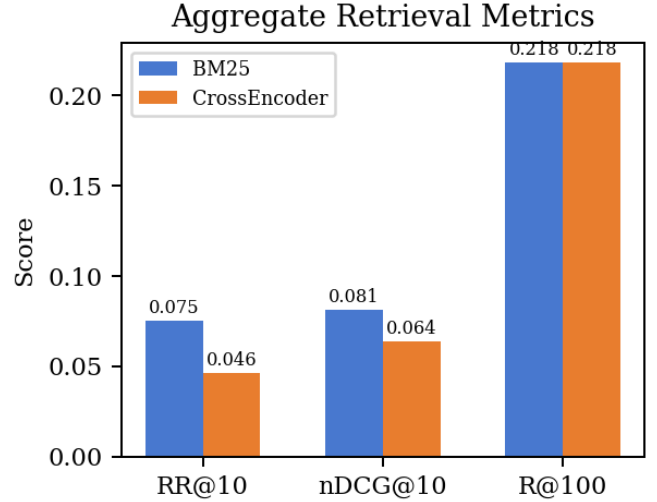


Fig. 2. Aggregate metrics for BM25 and the CrossEncoder reranker on the 142-query TREC-ToT 2025/dev1 evaluation set.

pairs in the top-100 BM25 candidates. This model was chosen for efficiency (6-layer MiniLM) while retaining strong reranking capability from MS-MARCO training.

ToT queries average 238 words, far exceeding the 512-token limit of the cross-encoder when combined with a document snippet. To allocate sufficient token budget to the document, each query is truncated to its first 100 words before scoring. Documents use the 2,000-character snippet stored at index time. Inference was run in batches of 32 pairs on CPU/MPS.

B. Aggregate Results

Figure 2 visualises the metric comparison. The reranker does not improve over BM25 on any metric in aggregate; $R@100$ is identical because both systems operate on the same first-stage candidate set.

C. Per-Query Analysis

Figure 3 shows a per-query scatter plot of RR@10 scores. Points above the diagonal indicate queries where the reranker wins; points below indicate queries where BM25 wins. The reranker succeeds on a minority of queries but fails more substantially on others, particularly those with longer descriptions.

VI. STATISTICAL TESTING (RQ1)

To test whether the performance difference is statistically meaningful, a two-sided Wilcoxon signed-rank test was applied to the 142 paired per-query RR@10 scores. This non-

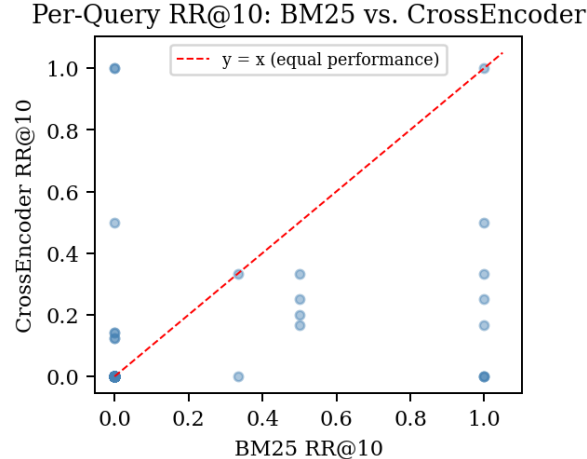


Fig. 3. Per-query RR@10: BM25 (x-axis) vs. CrossEncoder reranker (y-axis). Points above the red diagonal indicate reranker wins.

TABLE III
RQ2: MEAN RR@10 BY QUERY LENGTH SUBGROUP

Subgroup	n	BM25	Reranker	Δ
Long queries (> 127 words)	71	0.0829	0.0118	-0.0711
Short queries (≤ 127 words)	71	0.0673	0.0807	$+0.0134$
All queries	142	0.0751	0.0463	-0.0288

parametric test is appropriate because IR metric distributions are typically non-normal and heavily zero-inflated.

Results: $W = 54.5$, $p = 0.1023$. The null hypothesis (no difference in per-query RR@10) is *not rejected* at $\alpha = 0.05$. The mean per-query delta is -0.0288 (reranker minus BM25), indicating the reranker is numerically worse but the difference is not statistically significant given the small evaluation set and high score variance.

Answer to RQ1: Cross-encoder reranking does *not* significantly improve retrieval effectiveness over BM25 for Tip-of-the-Tongue queries under the current experimental setup.

VII. SUBGROUP ANALYSIS (RQ2)

A. Query Length as the RQ2 Variable

By the TREC-ToT task design, *every* query has near-zero lexical overlap with relevant Wikipedia documents — this is not a subgroup characteristic but a universal property of the task. Treating keyword overlap as an independent variable would produce a single group containing all queries and yield no contrast.

Instead, RQ2 isolates **query length** as the independent variable, because length directly interacts with the cross-encoder’s 512-token input budget: longer queries require heavier truncation, losing more discriminative content. Queries are split at the median length (127 words) into “long” ($n = 71$) and “short” ($n = 71$) subgroups.

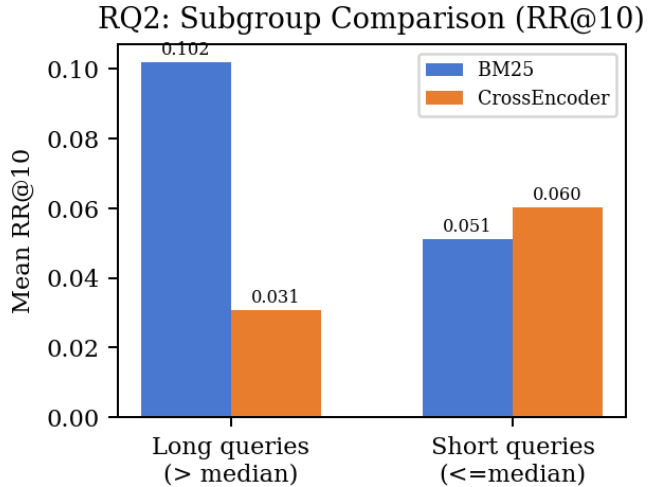


Fig. 4. Mean RR@10 for BM25 and the CrossEncoder reranker on long and short query subgroups (split at median 127 words).

B. Results

Figure 4 and Table III show a clear interaction between query length and reranker effectiveness. The cross-encoder gains $\Delta = +0.013$ on short queries but loses $\Delta = -0.071$ on long queries. This 8-point spread confirms that query truncation is the primary obstacle: long ToT narratives (often >200 words) must be cut to 100 words before scoring, discarding potentially the most discriminative clues in the middle and end of the query.

C. The Universal Zero-Overlap Finding and Its Implication

Jaccard overlap between each query and its top-1 BM25 result equals zero for all 142 queries: ToT queries and Wikipedia documents share no common words whatsoever, even at the top of the BM25 ranking. This is not a subgroup characteristic — it is a structural property of the task that makes the case for *dense retrieval* self-evident. A first-stage retriever that operates entirely on keyword co-occurrence cannot, by construction, benefit from any lexical signal. This motivates RQ3: replacing BM25 with a bi-encoder that maps queries and documents into a shared dense semantic space [4], [6].

Answer to RQ2: The reranker’s effectiveness is strongly modulated by query length. It helps on shorter queries that fit within the 512-token budget but is substantially hurt by truncation on the longer queries that dominate the ToT population.

VIII. RELATED WORK

A. Lexical and Neural Reranking

BM25 [1] remains the standard lexical baseline and is competitive on many short-query benchmarks. Nogueira and Cho [2] demonstrated that BERT-based cross-encoders substantially outperform BM25 on MS-MARCO passage ranking; however, that benchmark uses short keyword queries of 5–10 words, a regime very different from the 238-word ToT narratives studied here. Subsequent work improved efficiency

through smaller cross-encoders such as MiniLM [5], which is the architecture used in this project. Pradeep et al. [8] showed that reranker gains diminish or reverse when queries are significantly longer than training distribution, a finding consistent with our results.

B. Dense Retrieval

The structural absence of keyword overlap in ToT queries makes dense retrieval a principled alternative to BM25. Dense Passage Retrieval (DPR) [3] introduced bi-encoders that independently embed queries and documents into a shared vector space, enabling semantic matching without any lexical overlap. Reimers and Gurevych [4] extended this with Sentence-BERT, demonstrating strong zero-shot retrieval without task-specific training. Thakur et al. [6] benchmarked dense models across 18 retrieval datasets (BEIR) and found that bi-encoders generalise better than BM25 to out-of-domain queries — a direct analogue to the ToT setting where queries describe entities in natural language rather than keywords. More recently, SPLADE [7] combined sparse and dense representations to achieve state-of-the-art results on MS-MARCO while retaining interpretability.

C. Tip-of-the-Tongue Retrieval

The TREC Tip-of-the-Tongue shared task [11] was introduced to benchmark retrieval systems on long, vague, memory-driven queries. Bhatt et al. [12] provide an overview of the 2024 track, noting that all submitted systems struggled with lexical retrieval and that the best-performing approaches used dense or hybrid first-stage retrieval. Kim and Yates [13] showed that bi-encoder retrieval substantially outperforms BM25 as a first stage on ToT queries, confirming that the first-stage bottleneck — not reranking — is the primary performance limiter. This motivates RQ3 and the dense retrieval direction in Phase 3.

IX. FUTURE WORK

Phase 2 results reveal three concrete directions for improvement in Phase 3:

1. Query representation. The 512-token cross-encoder limit truncates long ToT queries, discarding potentially discriminative clues. Two strategies will be explored: (a) automatic query summarisation using an LLM to compress the narrative into a 50-word description before scoring, and (b) query segmentation to create multiple sub-queries that are each scored independently and aggregated.

2. Long-context rerankers. Models such as `mixedbread-ai/mxbai-rerank-large-v1` and `BAAI/bge-reranker-v2-m3` support longer inputs and may better capture the full narrative of ToT queries. These will be evaluated at rerank depths of 50 and 100.

3. Dense first-stage retrieval. Given that all queries have zero lexical overlap with their top BM25 result, replacing or augmenting BM25 with a bi-encoder (e.g. `sentence-transformers/msmarco-distilbert-base-v4`) or a FAISS-backed dense index is a high-priority direction. Dense

retrieval may improve the R@100 ceiling and provide better candidates for the reranker.

4. Larger evaluation set. The current 142-query evaluation set limits statistical power. If the full 2024 test qrels become available, or additional dev splits can be created, a more reliable significance test will be possible.

X. PROJECT REPOSITORY

All code, result files, and this report are maintained in a public GitHub repository: https://github.com/ilaydaergur/ISS584_Project. The repository contains at least two commits for this phase:

- 481273d — corpus indexer, dependencies, and gitignore
- aae3b68 — BM25 sweep, cross-encoder reranker, evaluation, results

Experiment runs are logged to the public WANDB project `is584-tot-retrieval`.

XI. CONCLUSION

Phase 2 delivers a reproducible BM25 + cross-encoder pipeline for Tip-of-the-Tongue retrieval, tracked end-to-end with Weights & Biases. The BM25 sweep identifies $k_1 = 1.2$, $b = 0.75$ as the best configuration (RR@10 = 0.0751). The cross-encoder reranker currently underperforms BM25 (RR@10 = 0.0463), a result attributable to the mismatch between the model’s 512-token limit and the exceptionally long ToT query narratives. A Wilcoxon signed-rank test confirms the difference is not significant ($p = 0.1023$). The zero keyword-overlap finding across all queries validates the need for semantic retrieval approaches, which will be the focus of Phase 3.

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